average stimulus triggered on two spikes separated by 5 ± 1 ms. The average stimulus triggered on a pair of spikes separated by 5 ms is not the same as the sum of the average stimuli for each spike separately.

Spike-triggered averages of other stimulus-dependent quantities can provide additional insight into neural encoding, for example, spike-triggered average autocorrelation functions. Obviously, spike-triggered averages of higher-order stimulus combinations can be considered as well.

1.4 Spike-Train Statistics

A complete description of the stochastic relationship between a stimulus and a response would require us to know the probabilities corresponding to every sequence of spikes that can be evoked by the stimulus. Spike times are continuous variables, and, as a result, the probability for a spike to occur at any precisely specified time is actually zero. To get a nonzero value, we must ask for the probability that a spike occurs within a specified interval, for example, the interval between times t and $t + \Delta t$. For small Δt, the probability of a spike falling in this interval is proportional to the size of the interval, Δt. A similar relation holds for any continuous stochastic variable z. The probability that z takes a value between z and $z + \Delta z$, for small Δz (strictly speaking, as $\Delta z \to 0$), is equal to $p[z]\Delta z$, where $p[z]$ is called a probability density.

Throughout this book, we use the notation $P[\,]$ to denote probabilities and $p[\,]$ to denote probability densities. We use the bracket notation $P[\,]$ generically for the probability of something occurring and also to denote a specific probability function. In the latter case, the notation $P(\,)$ would be more appropriate, but switching between square brackets and parentheses is confusing, so the reader will have to use the context to distinguish between these cases.

The probability of a spike sequence appearing is proportional to the probability density of spike times, $p[t_1, t_2, \ldots, t_n]$. In other words, the probability $P[t_1, t_2, \ldots, t_n]$ that a sequence of n spikes occurs with spike i falling between times t_i and $t_i + \Delta t$ for $i = 1, 2, \ldots, n$ is given in terms of this density by the relation $P[t_1, t_2, \ldots, t_n] = p[t_1, t_2, \ldots, t_n](\Delta t)^n$.

Unfortunately, the number of possible spike sequences is typically so large that determining or even roughly estimating all of their probabilities of occurrence is impossible. Instead, we must rely on some statistical model that allows us to estimate the probability of an arbitrary spike sequence occurring, given our knowledge of the responses actually recorded. The firing rate $\mathsf{r}(t)$ determines the probability of firing a single spike in a small interval around the time t, but $\mathsf{r}(t)$ is not, in general, sufficient information to predict the probabilities of spike sequences. For example, the probability of two spikes occurring together in a sequence is not necessarily equal to the product of the probabilities that they occur individually, because

the presence of one spike may effect the occurrence of the other. If, however, the probability of generating an action potential is independent of the presence or timing of other spikes (i.e., if the spikes are statistically independent) the firing rate is all that is needed to compute the probabilities for all possible action potential sequences.

point process

A stochastic process that generates a sequence of events, such as action potentials, is called a point process. In general, the probability of an event occurring at any given time could depend on the entire history of preceding events. If this dependence extends only to the immediately preceding event, so that the intervals between successive events are independent, the point process is called a renewal process. If there is no dependence at all on preceding events, so that the events themselves are statistically independent, we have a Poisson process. The Poisson process provides an extremely useful approximation of stochastic neuronal firing. To make the presentation easier to follow, we separate two cases, the homogeneous Poisson process, for which the firing rate is constant over time, and the inhomogeneous Poisson process, which involves a time-dependent firing rate.

renewal process

Poisson process

The Homogeneous Poisson Process

We denote the firing rate for a homogeneous Poisson process by $\mathsf{r}(t) = \mathsf{r}$ because it is independent of time. When the firing rate is constant, the Poisson process generates every sequence of n spikes over a fixed time interval with equal probability. As a result, the probability $P[t_1, t_2, \ldots, t_n]$ can be expressed in terms of another probability function $P_T[n]$, which is the probability that an arbitrary sequence of exactly n spikes occurs within a trial of duration T. Assuming that the spike times are ordered so that $0 \leq t_1 \leq t_2 \leq \ldots \leq t_n \leq T$, the relationship is

$$P[t_1, t_2, \ldots, t_n] = n! P_T[n] \left(\frac{\Delta t}{T}\right)^n . \tag{1.26}$$

This relationship is a special case of equation 1.37 derived below.

To compute $P_T[n]$, we divide the time T into M bins of size $\Delta t = T/M$. We can assume that Δt is small enough so that we never get two spikes within any one bin because, at the end of the calculation, we take the limit $\Delta t \to 0$. $P_T[n]$ is the product of three factors: the probability of generating n spikes within a specified set of the M bins, the probability of not generating spikes in the remaining $M - n$ bins, and a combinatorial factor equal to the number of ways of putting n spikes into M bins. The probability of a spike occurring in one specific bin is $\mathsf{r}\Delta t$, and the probability of n spikes appearing in n specific bins is $(\mathsf{r}\Delta t)^n$. Similarly, the probability of not having a spike in a given bin is $(1 - \mathsf{r}\Delta t)$, so the probability of having the remaining $M - n$ bins without any spikes in them is $(1 - \mathsf{r}\Delta t)^{M-n}$. Finally, the number of ways of putting n spikes into M bins is given by the

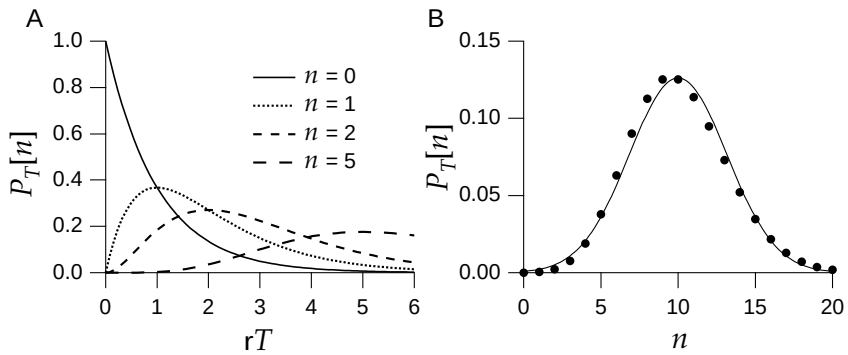

Figure 1.11 (A) The probability that a homogeneous Poisson process generates n spikes in a time period of duration T plotted for n = 0, 1, 2, and 5. The probability is plotted as function of the rate times the duration of the interval, rT, to make the plot applicable for any rate. (B) The probability of finding n spikes during a time period for which $\mathsf{r}T = 10$ (dots) compared with a Gaussian distribution with mean and variance equal to 10 (line).

binomial coefficient $M!/(M-n)!n!$. Putting all these factors together, we find

$$P_T[n] = \lim_{\Delta t \to 0} \frac{M!}{(M-n)!n!} (\mathsf{r}\Delta t)^n (1 - \mathsf{r}\Delta t)^{M-n} . \tag{1.27}$$

To take the limit, we note that as $\Delta t \to 0$, M grows without bound because $M\Delta t = T$. Because n is fixed, we can write $M - n \approx M = T/\Delta t$. Using this approximation and defining $\epsilon = -\mathsf{r}\Delta t$, we find that

$$\lim_{\Delta t \to 0} (1 - \mathsf{r}\Delta t)^{M-n} = \lim_{\epsilon \to 0} \left((1+\epsilon)^{1/\epsilon}\right)^{-\mathsf{r}T} = e^{-\mathsf{r}T} = \exp(-\mathsf{r}T) \tag{1.28}$$

because $\lim_{\epsilon \to 0}(1+\epsilon)^{1/\epsilon}$ is, by definition, $e = \exp(1)$. For large M, $M!/(M-n)! \approx M^n = (T/\Delta t)^n$, so

$$P_T[n] = \frac{(\mathsf{r}T)^n}{n!} \exp(-\mathsf{r}T) . \tag{1.29}$$

Poisson distribution

This is called the Poisson distribution. The probabilities $P_T[n]$, for a few n values, are plotted as a function of rT in figure 1.11A. Note that as n increases, the probability reaches its maximum at larger T values and that large n values are more likely than small ones for large T. Figure 1.11B shows the probabilities of various numbers of spikes occurring when the average number of spikes is 10. For large rT, which corresponds to a large expected number of spikes, the Poisson distribution approaches a Gaussian distribution with mean and variance equal to rT. Figure 1.11B shows that this approximation is already quite good for rT = 10.

We can compute the variance of spike counts produced by a Poisson process from the probabilities in equation 1.29. For spikes counted over an interval of duration T, the variance of the spike count (derived in appendix B) is

$$\sigma_n^2 = \langle n^2 \rangle - \langle n \rangle^2 = \mathsf{r}T . \tag{1.30}$$

Thus the variance and mean of the spike count are equal. The ratio of these two quantities, $\sigma_n^2/\langle n\rangle$, is called the Fano factor and takes the value 1 for a homogeneous Poisson process, independent of the time interval T.

Fano factor

The probability density of time intervals between adjacent spikes is called the interspike interval distribution, and it is a useful statistic for characterizing spiking patterns. Suppose that a spike occurs at a time t_i for some value of i. The probability of a homogeneous Poisson process generating the next spike somewhere in the interval $t_i + \tau \leq t_{i+1} < t_i + \tau + \Delta t$, for small Δt, is the probability that no spike is fired for a time τ, times the probability, $\mathsf{r}\Delta t$, of generating a spike within the following small interval Δt. From equation 1.29, with $n = 0$, the probability of not firing a spike for period τ is $\exp(-\mathsf{r}\tau)$, so the probability of an interspike interval falling between τ and $\tau + \Delta t$ is

interspike interval distribution

$$P[\tau \leq t_{i+1} - t_i < \tau + \Delta t] = \mathsf{r}\Delta t \exp(-\mathsf{r}\tau)\,. \tag{1.31}$$

The probability density of interspike intervals is, by definition, this probability with the factor Δt removed. Thus, the interspike interval distribution for a homogeneous Poisson spike train is an exponential. The most likely interspike intervals are short ones, and long intervals have a probability that falls exponentially as a function of their duration.

From the interspike interval distribution of a homogeneous Poisson spike train, we can compute the mean interspike interval,

$$\langle\tau\rangle = \int_0^\infty d\tau\, \tau \mathsf{r} \exp(-\mathsf{r}\tau) = \frac{1}{\mathsf{r}}\,, \tag{1.32}$$

and the variance of the interspike intervals,

$$\sigma_\tau^2 = \int_0^\infty d\tau\, \tau^2 \mathsf{r} \exp(-\mathsf{r}\tau) - \langle\tau\rangle^2 = \frac{1}{\mathsf{r}^2}\,. \tag{1.33}$$

The ratio of the standard deviation to the mean is called the coefficient of variation,

coefficient of variation C_V

$$C_V = \frac{\sigma_\tau}{\langle\tau\rangle}\,, \tag{1.34}$$

and it takes the value 1 for a homogeneous Poisson process. This is a necessary, though not sufficient, condition to identify a Poisson spike train. Recall that the Fano factor for a Poisson process is also 1. For any renewal process, the Fano factor evaluated over long time intervals approaches the value C_V^2.

The Spike-Train Autocorrelation Function

The spike interval distribution measures the distribution of times between successive action potentials in a train. It is useful to generalize this concept and determine the distribution of times between any two spikes in

a train. This is called the spike-train autocorrelation function, and it is particularly useful for detecting patterns in spike trains, most notably oscillations. The spike-train autocorrelation function is the autocorrelation of the neural response function of equation 1.1 with its average over time and trials subtracted out. The time average of the neural response function, from equation 1.4, is the spike-count rate r, and the trial average of this quantity is $\langle r \rangle = \langle n \rangle / T$. Thus, the spike-train autocorrelation function is

spike-train autocorrelation function $Q_{\rho\rho}$

$$Q_{\rho\rho}(\tau) = \frac{1}{T}\int_0^T dt\ \langle (\rho(t) - \langle r \rangle)\,(\rho(t+\tau) - \langle r \rangle)\rangle\ . \tag{1.35}$$

Because the average is subtracted from the neural response function in this expression, $Q_{\rho\rho}$ should really be called an autocovariance, not an autocorrelation, but in practice it isn't.

The spike-train autocorrelation function is constructed from data in the form of a histogram by dividing time into bins. The value of the histogram for a bin labeled with a positive or negative integer m is computed by determining the number of the times that any two spikes in the train are separated by a time interval lying between $(m - 1/2)\,\Delta t$ and $(m + 1/2)\,\Delta t$ with Δt the bin size. This includes all pairings, even between a spike and itself. We call this number N_m. If the intervals between the n^2 spike pairs in the train were uniformly distributed over the range from 0 to T, there would be $n^2\,\Delta t/T$ intervals in each bin. This uniform term is removed from the autocorrelation histogram by subtracting $n^2\,\Delta t/T$ from N_m for all m. The spike-train autocorrelation histogram is then defined by dividing the resulting numbers by T, so the value of the histogram in bin m is $H_m = N_m/T - n^2\Delta t/T^2$. For small bin sizes, the $m = 0$ term in the histogram counts the average number of spikes, that is $N_0 = \langle n \rangle$ and in the limit $\Delta t \to 0$, $H_0 = \langle n \rangle / T$ is the average firing rate $\langle r \rangle$. Because other bins have H_m of order Δt, the large $m = 0$ term is often removed from histogram plots. The spike-train autocorrelation function is defined as $H_m/\Delta t$ in the limit $\Delta t \to 0$, and it has the units of a firing rate squared. In this limit, the $m = 0$ bin becomes a δ function, $H_0/\Delta t \to \langle r \rangle \delta(\tau)$.

As we have seen, the distribution of interspike intervals for adjacent spikes in a homogeneous Poisson spike train is exponential (equation 1.31). By contrast, the intervals between any two spikes (not necessarily adjacent) in such a train are uniformly distributed. As a result, the subtraction procedure outlined above gives $H_m = 0$ for all bins except for the $m = 0$ bin that contains the contribution of the zero intervals between spikes and themselves. The autocorrelation function for a Poisson spike train generated at a constant rate $\langle r \rangle = \mathsf{r}$ is thus

$$Q_{\rho\rho}(\tau) = \mathsf{r}\delta(\tau)\ . \tag{1.36}$$

cross-correlation function

A cross-correlation function between spike trains from two different neurons can be defined by analogy with the autocorrelation function by determining the distribution of intervals between pairs of spikes, one taken

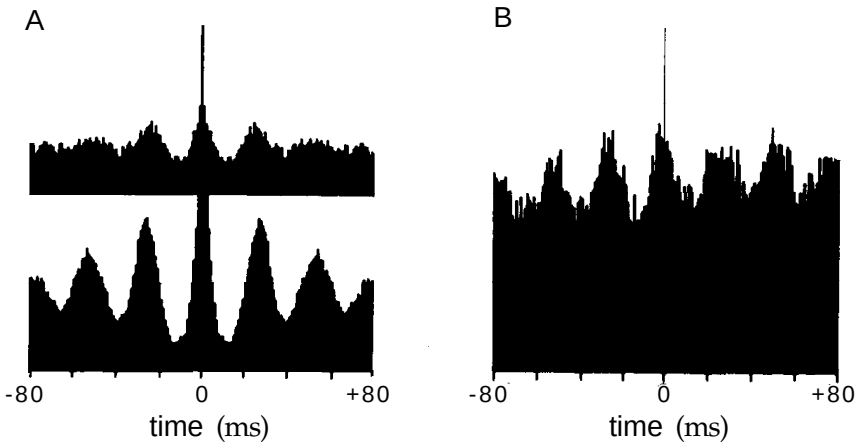

Figure 1.12 Autocorrelation and cross-correlation histograms for neurons in the primary visual cortex of a cat. (A) Autocorrelation histograms for neurons recorded in the right (upper) and left (lower) hemispheres show a periodic pattern indicating oscillations at about 40 Hz. The lower diagram indicates stronger oscillations in the left hemisphere. (B) The cross-correlation histogram for these two neurons shows that their oscillations are synchronized with little time delay. (Adapted from Engel et al., 1991.)

from each train. The spike-train autocorrelation function is an even function of τ, $Q_{\rho\rho}(\tau) = Q_{\rho\rho}(-\tau)$, but the cross-correlation function is not necessarily even. A peak at zero interval in a cross-correlation function signifies that the two neurons are firing synchronously. Asymmetric shifts in this peak away from 0 result from fixed delays between the firing of the two neurons, and they indicate nonsynchronous but phase-locked firing. Periodic structure in either an autocorrelation or a cross-correlation function or histogram indicates that the firing probability oscillates. Such periodic structure is seen in the histograms of figure 1.12, showing 40 Hz oscillations in neurons of cat primary visual cortex that are roughly synchronized between the two cerebral hemispheres.

The Inhomogeneous Poisson Process

When the firing rate depends on time, different sequences of n spikes occur with different probabilities, and $p[t_1, t_2, \ldots, t_n]$ depends on the spike times. Because spikes are still generated independently by an inhomogeneous Poisson process, their times enter into $p[t_1, t_2, \ldots, t_n]$ only through the time-dependent firing rate $\mathsf{r}(t)$. Assuming, as before, that the spike times are ordered $0 \le t_1 \le t_2 \le \ldots \le t_n \le T$, the probability density for n spike times (derived in appendix C) is

$$p[t_1, t_2, \ldots, t_n] = \exp\left(-\int_0^T dt\, \mathsf{r}(t)\right) \prod_{i=1}^{n} \mathsf{r}(t_i)\,. \tag{1.37}$$

This result applies if the spike times have been written in temporal order. If the spike times are not ordered, so that, for example, we are interested in the probability density for any spike occurring at the time t_1, not necessarily the first spike, this expression should be divided by a factor of $n!$ to account for the number of different possible orderings of spike times.

The Poisson Spike Generator

Spike sequences can be simulated by using some estimate of the firing rate, $\mathsf{r}_{\text{est}}(t)$, predicted from knowledge of the stimulus, to drive a Poisson process. A simple procedure for generating spikes in a computer program is based on the fact that the estimated probability of firing a spike during a short interval of duration Δt is $\mathsf{r}_{\text{est}}(t)\,\Delta t$. The program progresses through time in small steps of size Δt and generates, at each time step, a random number x_{rand} chosen uniformly in the range between 0 and 1. If $\mathsf{r}_{\text{est}}(t)\,\Delta t > x_{\text{rand}}$ at that time step, a spike is fired; otherwise it is not.

For a constant firing rate, it is faster to compute spike times t_i for $i = 1, 2, \ldots n$ iteratively by generating interspike intervals from an exponential probability density (equation 1.31). If x_{rand} is uniformly distributed over the range between 0 and 1, the negative of its logarithm is exponentially distributed. Thus, we can generate spike times iteratively from the formula $t_{i+1} = t_i - \ln(x_{\text{rand}})/\mathsf{r}$. Unlike the algorithm discussed in the previous paragraph, this method works only for constant firing rates. However, it can be extended to time-dependent rates by using a procedure called rejection sampling or spike thinning. The thinning technique requires a bound $\mathsf{r}_{\max}$ on the estimated firing rate such that $\mathsf{r}_{\text{est}}(t) \le \mathsf{r}_{\max}$ at all times. We first generate a spike sequence corresponding to the constant rate $\mathsf{r}_{\max}$ by iterating the rule $t_{i+1} = t_i - \ln(x_{\text{rand}})/\mathsf{r}_{\max}$. The spikes are then thinned by generating another x_{rand} for each i and removing the spike at time t_i from the train if $\mathsf{r}_{\text{est}}(t_i)/\mathsf{r}_{\max} < x_{\text{rand}}$. If $\mathsf{r}_{\text{est}}(t_i)/\mathsf{r}_{\max} \ge x_{\text{rand}}$, spike i is retained. Thinning corrects for the difference between the estimated time-dependent rate and the maximum rate.

Figure 1.13 shows an example of a model of an orientation-selective V1 neuron constructed in this way. In this model, the estimated firing rate is determined from the response tuning curve of figure 1.5B,

$$\mathsf{r}_{\text{est}}(t) = f(s(t)) = r_{\max} \exp\left(-\frac{1}{2}\left(\frac{s(t) - s_{\max}}{\sigma_f}\right)^2\right). \tag{1.38}$$

This is an extremely simplified model of response dynamics, because the firing rate at any given time depends only on the value of the stimulus at that instant of time and not on its recent history. Models that allow for a dependence of firing rate on stimulus history are discussed in chapter 2. In figure 1.13, the orientation angle increases in a sequence of steps. The firing rate follows these changes, and the Poisson process generates an irregular firing pattern that reflects the underlying rate but varies from trial to trial.

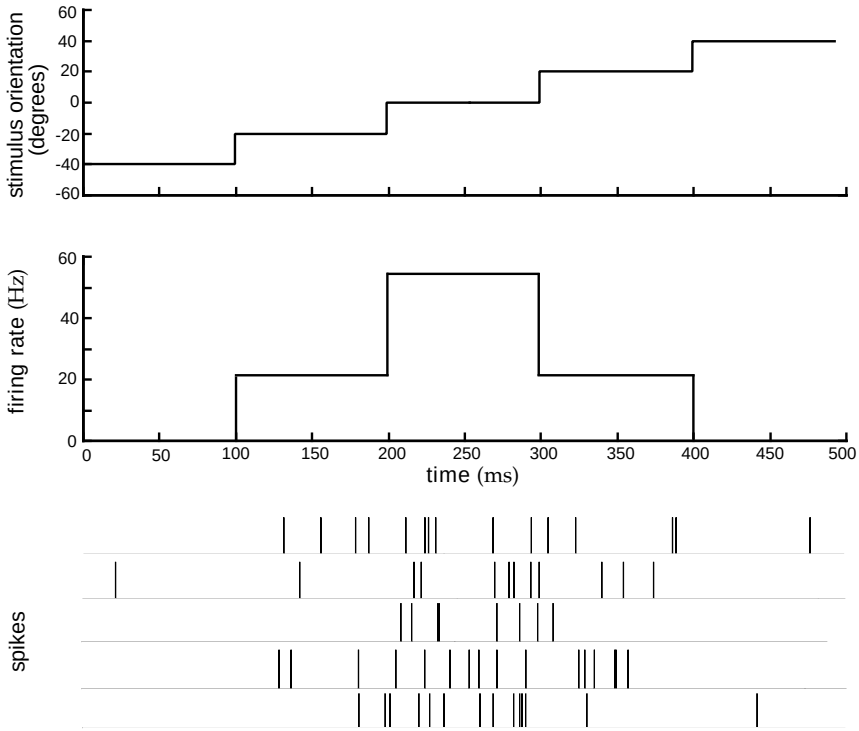

Figure 1.13 Model of an orientation-selective neuron. The orientation angle (top panel) was increased from an initial value of -40° by 20° every 100 ms. The firing rate (middle panel) was used to generate spikes (bottom panel) using a Poisson spike generator. The bottom panel shows spike sequences generated on five different trials.

Certain features of neuronal firing violate the independence assumption that forms the basis of the Poisson model, at least if a constant firing rate is used. We have already mentioned the absolute and relative refractory periods, which are periods of time following the generation of an action potential when the probability of a spike occurring is greatly or somewhat reduced. Frequently, these are most prominent features of real neuronal spike trains that are not captured by a Poisson model. Refractory effects can be incorporated into a Poisson model of spike generation by setting the firing rate to 0 immediately after a spike is fired, and then letting it return to its predicted value according to some dynamic rule such as an exponential recovery.

Comparison with Data

The Poisson process is simple and useful, but does it match data on neural response variability? To address this question, we examine Fano factors, interspike interval distributions, and coefficients of variation.

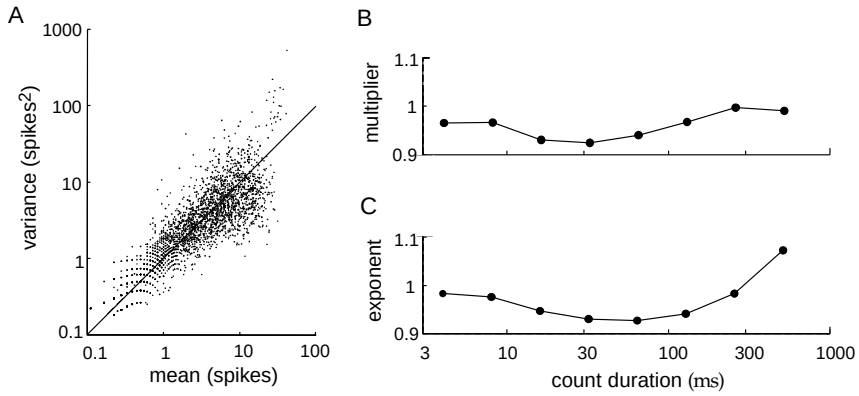

Figure 1.14 Variability of MT neurons in alert macaque monkeys responding to moving visual images. (A) Variance of the spike counts for a 256 ms counting period plotted against the mean spike count. The straight line is the prediction of the Poisson model. Data are from 94 cells recorded under a variety of stimulus conditions. (B) The multiplier A in the relationship between spike-count variance and mean as a function of the duration of the counting interval. (C) The exponent B in this relation as a function of the duration of the counting interval. (Adapted from O'Keefe et al., 1997.)

The Fano factor describes the relationship between the mean spike count over a given interval and the spike-count variance. Mean spike counts $\langle n \rangle$ and variances σ_n^2 from a wide variety of neuronal recordings have been fitted to the equation $\sigma_n^2 = A\langle n \rangle^B$, and the multiplier A and exponent B have been determined. The values of both A and B typically lie between 1.0 and 1.5. Because the Poisson model predicts $A = B = 1$, this indicates that the data show a higher degree of variability than the Poisson model would predict. However, many of these experiments involve anesthetized animals, and it is known that response variability is higher in anesthetized than in alert animals.

area MT

Figure 1.14 shows data for spike-count means and variances extracted from recordings of MT neurons in alert macaque monkeys using a number of different stimuli. The MT (medial temporal) area is a visual region of the primate cortex where many neurons are sensitive to image motion. The individual means and variances are scattered in figure 1.14A, but they cluster around the diagonal which is the Poisson prediction. Similarly, the results show A and B values close to 1, the Poisson values (figure 1.14B). Of course, many neural responses cannot be described by Poisson statistics, but it is reassuring to see a case where the Poisson model seems a reasonable approximation. As mentioned previously, when spike trains are not described very accurately by a Poisson model, refractory effects are often the primary reason.

Interspike interval distributions are extracted from data as interspike interval histograms by counting the number of intervals falling in discrete time bins. Figure 1.15A presents an example from the responses of a nonbursting cell in area MT of a monkey in response to images consisting of

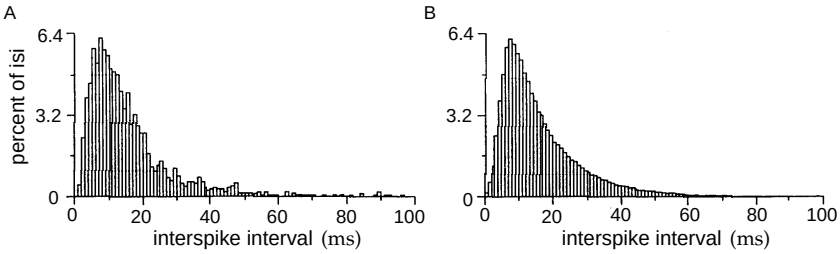

Figure 1.15 (A) Interspike interval distribution from an MT neuron responding to a moving, random-dot image. The probability of interspike intervals falling into the different bins, expressed as a percentage, is plotted against interspike interval. (B) Interspike interval histogram generated from a Poisson model with a stochastic refractory period. (Adapted from Bair et al., 1994.)

randomly moving dots with a variable amount of coherence imposed on their motion (see chapter 3 for a more detailed description). For interspike intervals longer than about 10 ms, the shape of this histogram is exponential, in agreement with equation 1.31. However, for shorter intervals there is a discrepancy. While the homogeneous Poisson distribution of equation 1.31 rises for short interspike intervals, the experimental results show a rapid decrease. This is the result of refractoriness making short interspike intervals less likely than the Poisson model would predict. Data on interspike intervals can be fitted more accurately by a gamma distribution,

gamma distribution

$$p[\tau] = \frac{r(r\tau)^k \exp(-r\tau)}{k!} \tag{1.39}$$

with $k > 0$, than by the exponential distribution of the Poisson model, which has $k = 0$.

Figure 1.15B shows a theoretical histogram obtained by adding a refractory period of variable duration to the Poisson model. Spiking was prohibited during the refractory period, and then was described once again by a homogeneous Poisson process. The refractory period was randomly chosen from a Gaussian distribution with a mean of 5 ms and a standard deviation of 2 ms (only random draws that generated positive refractory periods were included). The resulting interspike interval distribution of figure 1.15B agrees quite well with the data.

C_V values extracted from the spike trains of neurons recorded in monkeys from area MT and primary visual cortex (V1) are shown in figure 1.16. The data have been divided into groups based on the mean interspike interval, and the coefficient of variation is plotted as a function of this mean interval, equivalent to $1/\langle r \rangle$. Except for short mean interspike intervals, the values are near 1, although they tend to cluster slightly lower than 1, the Poisson value. The small C_V values for short interspike intervals are due to the refractory period. The solid curve is the prediction of a Poisson model with refractoriness.

The Poisson model with refractoriness provides a reasonably good description of a significant amount of data, especially considering its sim-

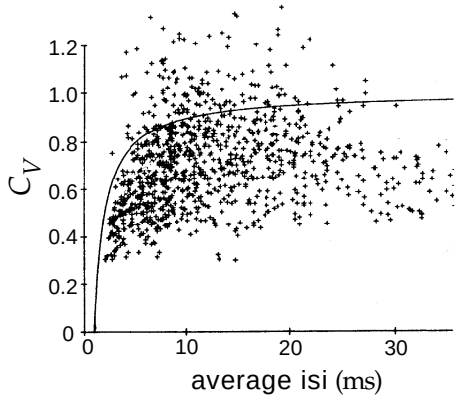

Figure 1.16 Coefficients of variation for a large number of V1 and MT neurons plotted as a function of mean interspike interval. The solid curve is the result of a Poisson model with a refractory period. (Adapted from Softky and Koch, 1992.)

plicity. However, there are cases in which the accuracy in the timing and numbers of spikes fired by a neuron is considerably higher than would be implied by Poisson statistics. Furthermore, even when it successfully describes data, the Poisson model does not provide a mechanistic explanation of neuronal response variability. Spike generation, by itself, is highly reliable in real neurons. Figure 1.17 compares the response of V1 cells to constant current injection in vivo and in vitro. The in vitro response is a regular and reproducible spike train (left panel). The same current injection paradigm applied in vivo produces a highly irregular pattern of firing (center panel) similar to the response to a moving bar stimulus (right panel). Although some of the basic statistical properties of firing variability may be captured by the Poisson model of spike generation, the spike-generating mechanism itself in real neurons is clearly not responsible for the variability. We explore ideas about possible sources of spike-train variability in chapter 5.

Some neurons fire action potentials in clusters or bursts of spikes that cannot be described by a Poisson process with a fixed rate. Bursting can be included in a Poisson model by allowing the firing rate to fluctuate in order to describe the high rate of firing during a burst. Sometimes the distribution of bursts themselves can be described by a Poisson process (such a doubly stochastic process is called a Cox process).

1.5 The Neural Code

The nature of the neural code is a topic of intense debate within the neuroscience community. Much of the discussion has focused on whether neurons use rate coding or temporal coding, often without a clear definition of what these terms mean. We feel that the central issue in neural coding is whether individual action potentials and individual neurons encode inde-